A Hybrid Neurogenetic Approach for Stock Forecasting

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Abstract (1/2)

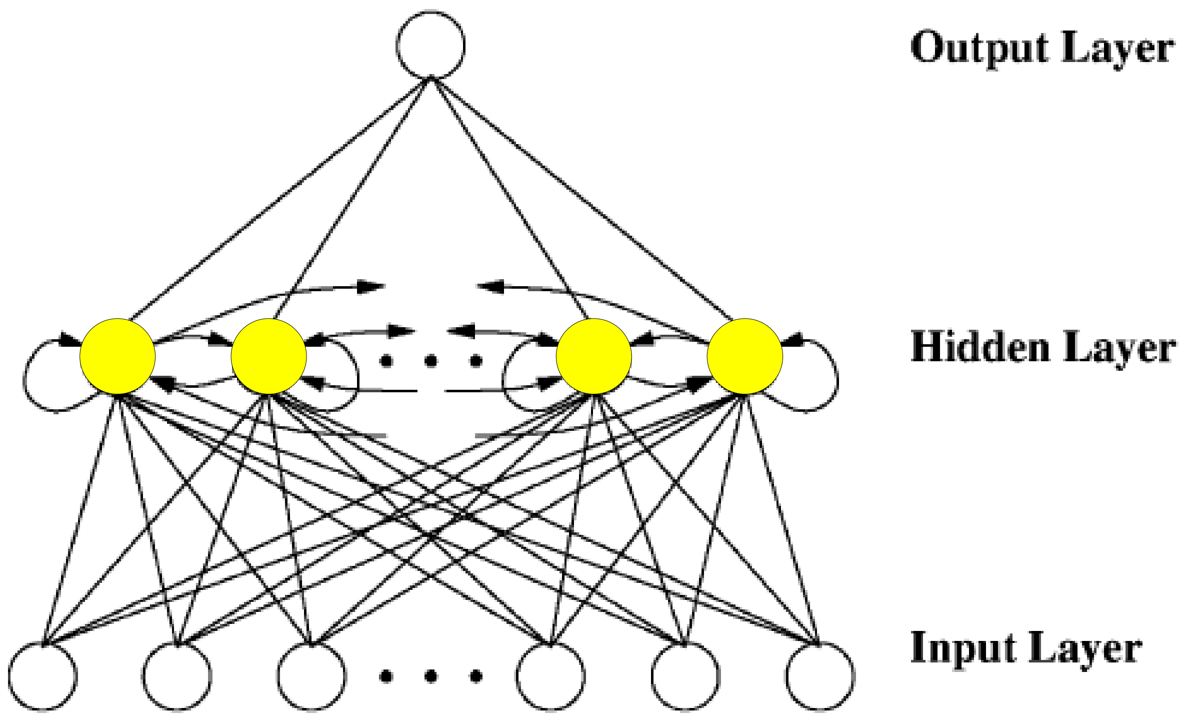
- In this paper, we propose a hybrid neurogenetic system for stock trading. A recurrent neural network (NN) having one hidden layer is used for the prediction model. The input features are generated from a number of technical indicators being used by financial experts. The genetic algorithm (GA) optimizes the NN's weights under a 2-D encoding and crossover. We devised a context-based ensemble method of NNs which dynamically changes on the basis of the test day's context.

Abstract (2/2)

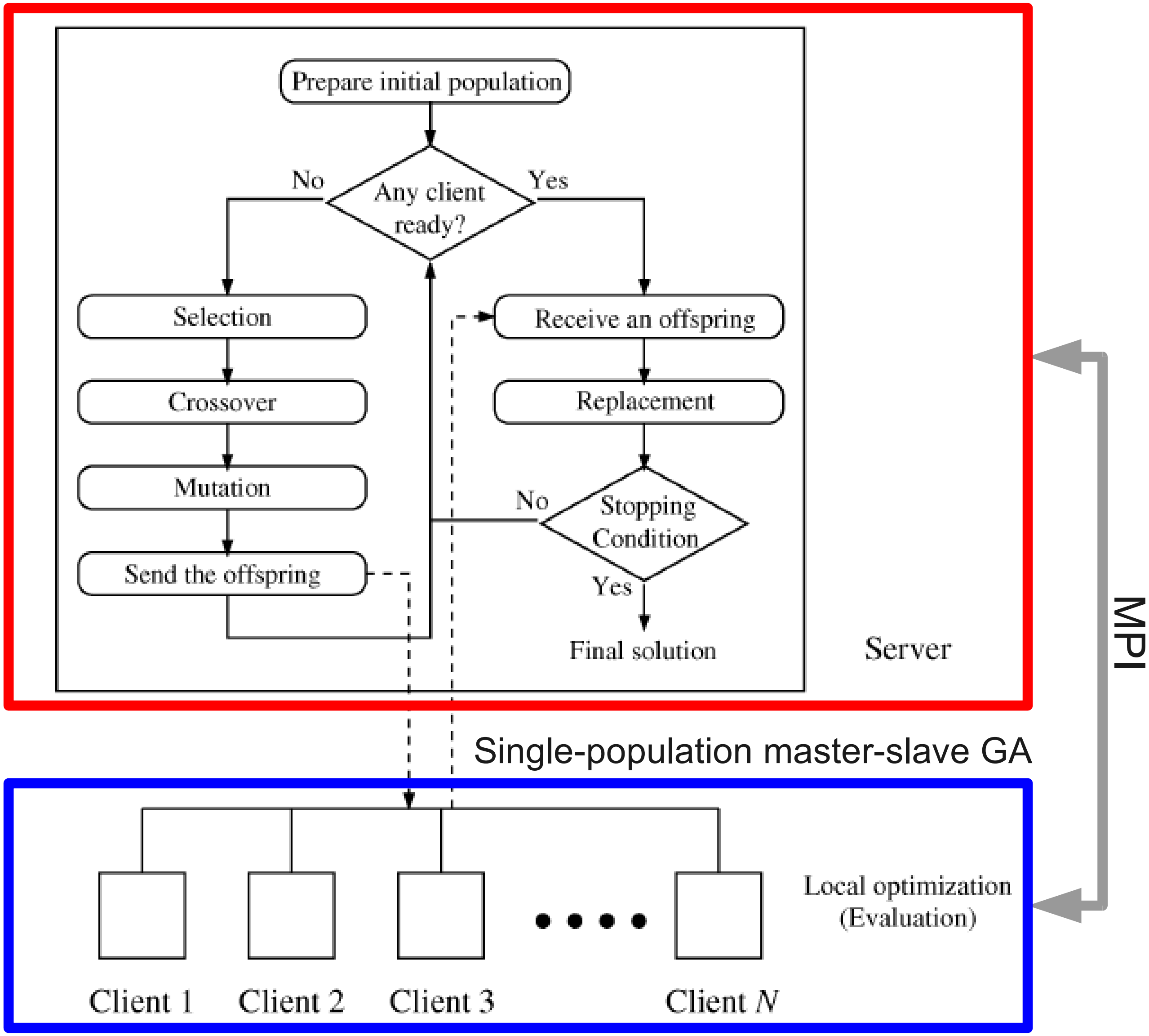
- To reduce the time in processing mass data, we parallelized the GA on a Linux cluster system using message passing interface. We tested the proposed method with 36 companies in NYSE and NASDAQ for 13 years from 1992 to 2004. The neurogenetic hybrid showed notable improvement on the average over the buy-and-hold strategy and the context-based ensemble further improved the results. We also observed that some companies were more predictable than others, which implies that the proposed neurogenetic hybrid can be used for financial portfolio construction.

System framework (1/2)

- Recurrent NN architectural



System framework (2/2)



Invest strategy

if (signal is $SELL$) {
$$C_{t+1} \leftarrow C_t + min(B, S_t) \times (1 - T)$$
$$S_{t+1} \leftarrow S_t - min(B, S_t)$$
}
if (signal is BUY) {
$$C_{t+1} \leftarrow C_t - min(B, C_t)$$
$$S_{t+1} \leftarrow S_t + min(B, C_t)$$
}
$$S_{t+1} \leftarrow S_t \times \frac{x_{t+1}}{x_t}$$

C: cash balance
S: stock
B: upper-bound of stock trade
T: one-way transaction cost

x: the closing price
t: the day

signal:
SELL: x(t+1)-x(t) < threshold
BUY: x(t+1)-x(t) > threshold

That is, we object to the property ratio *P*:

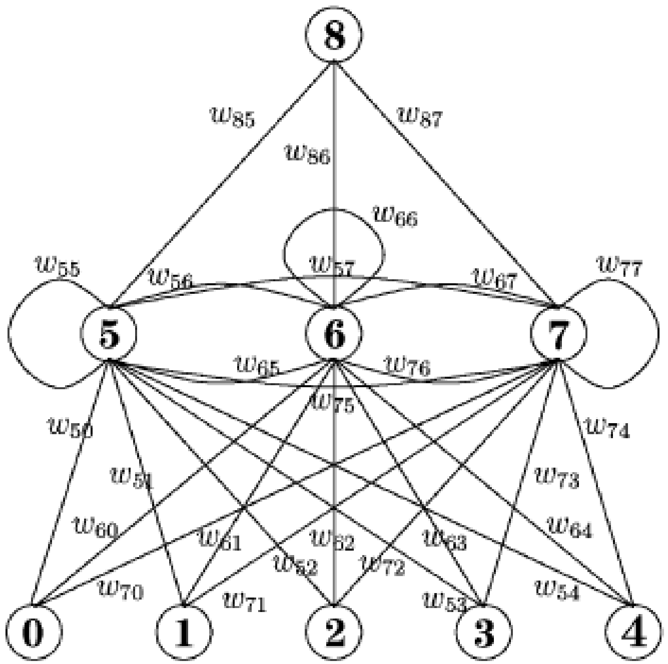
$$P = \frac{C_N + S_N}{C_1 + S_1}$$

Features

- There are 75 features

$$
\begin{aligned}
X_1 &= (x(t) - x(t-1))/x(t-1) \\
X_2 &= (x(t) - x_l(t))/x_h(t) - x_l(t) \\
X_3 &= (MA(t) - MA(t-1))/MA(t-1) \\
X_4 &= (MA_S(t) - MA_L(t))/MA_L(t) \\
X_5 &= (x(t) - MA(t))/MA(t) \\
X_6 &= x(t) - min(x(t), x(t-1), \ldots, x(t-5)) \\
X_7 &= x(t) - max(x(t), x(t-1), \ldots, x(t-5)) \\
X_8 &= \text{\# of days since the last golden-cross} \\
X_9 &= \text{\# of days since the last dead-cross} \\
X_{10} &= \text{the profit while the stock has risen or fallen continuously} \\
X_{11} &= \text{\# of days for which the stock has risen or fallen continuously} \\
X_{12} &= MACD(t) \\
X_{13} &= RSI(t) \\
X_{14} &= \%K(t) \\
X_{15} &= \%D(t) \\
X_{16} &= \%K(t) - \%K(t-1) \\
X_{17} &= \%D(t) - \%D(t-1)
\end{aligned}
$$

ANN with GA (2/2)



A recurrent neural network

Input: features

Output: the change ratio $(x_{t+1} - x_t)/x_t$

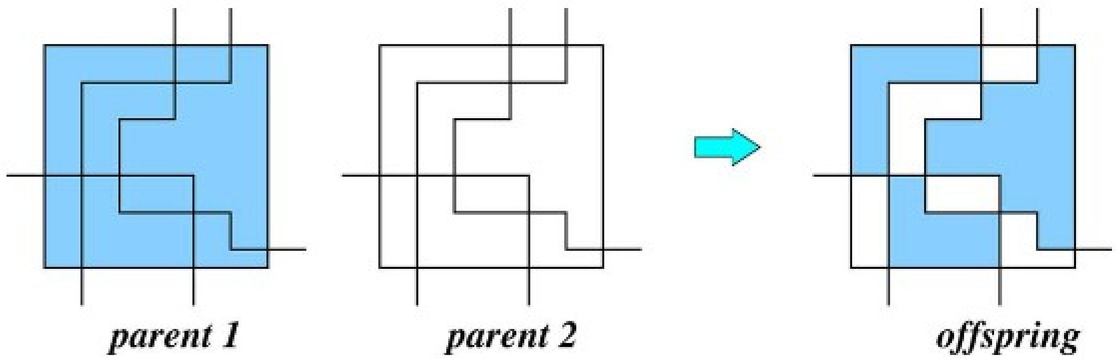
	input					*hidden*			*output*
	0	1	2	3	4	5	6	7	8
5	w_{50}	w_{51}	w_{52}	w_{53}	w_{54}	w_{55}	w_{56}	w_{57}	w_{85}
6	w_{60}	w_{61}	w_{62}	w_{63}	w_{64}	w_{65}	w_{66}	w_{67}	w_{86}
7	w_{70}	w_{71}	w_{72}	w_{73}	w_{74}	w_{75}	w_{76}	w_{77}	w_{87}

Corresponding 2-D chromosome

There are 3 type of signals: *BUY, SELL and KEEP*,
where the threshold is 0.005 and -0.005.

ANN with GA (2/2)

- Selection, crossover and mutation



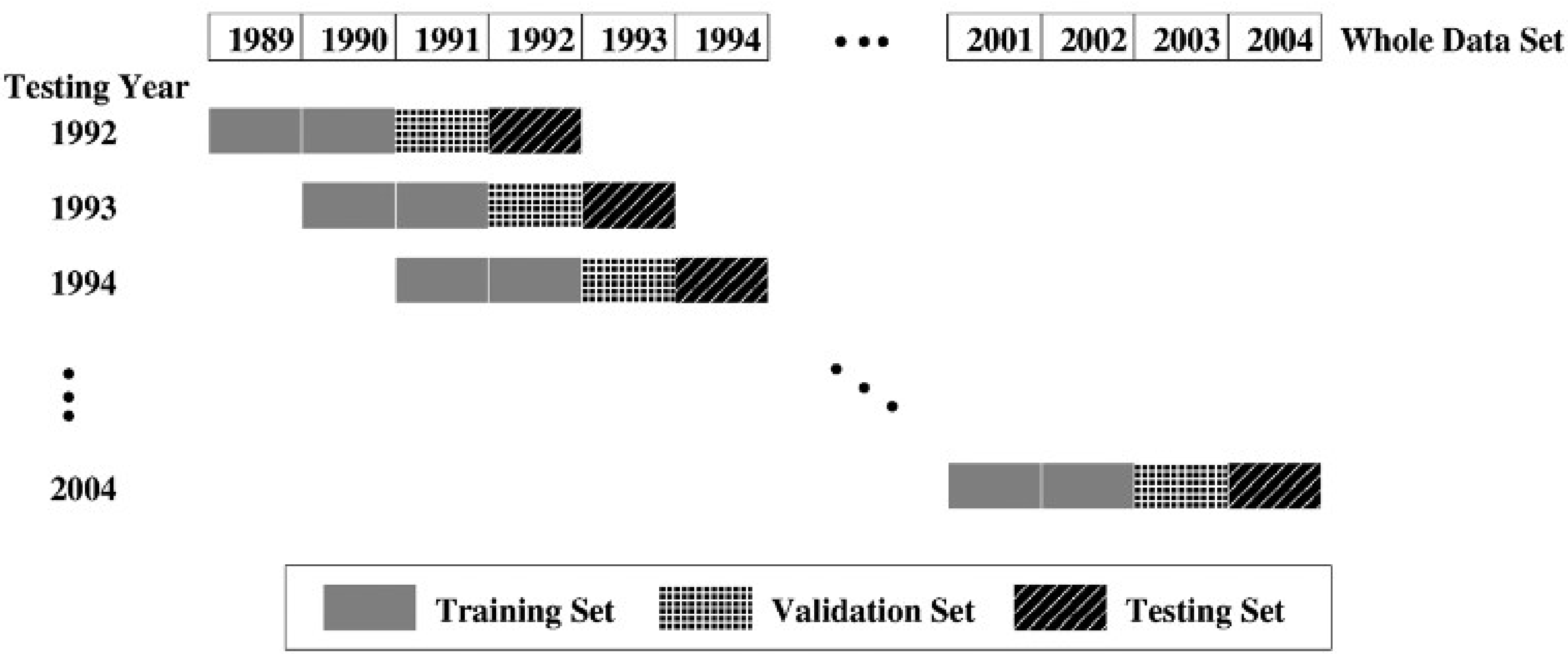
- Fitness: $F_k = P_k - P_w + (P_b - P_w)/3$

where

F_k	fitness of chromosome k;
P_k	final property ratio of chromosome k
b, w	the indices of the best and the worst chromosomes in the population, respectively.

Walk-forward testing scheme

- Data set: 36 companies of 13 years



Experimental result (1/10)

- Ensemble of NNs:
  - The method is based on the fact that a solution with the smallest training error does not necessarily guarantee the most generalized one.
  - Not like traditional, they are not fixed member size but using *k*-means algorithm to clustering.

Experimental result (2/10)

- Type:
  - Winner: using the best one
  - I-ensemble: using the best set
  - A-ensemble: make decision of the average from the outputs of them
  - M-ensemble: like A-type but using the majority opinion
- Signal:
  - BUY: by the next two days are all UP
  - SELL: by the next two days are all DOWN
  - KEEP(NC): otherwise

Experimental result (3/10)

RELATIVE PERFORMANCE OF RNN AND GA OVER THE BUY-AND-HOLD (36 COMPANIES AND $T = 0$)

Symbols	Market State			RNN		GA	
	Up	*Down*	*NC*	*Better*	*Worse*	*Better*	*Worse*
AA	8	3	2	5	4	**8**	**1**
AXP	10	1	2	**2**	**3**	2	5
AYP	8	4	1	4	2	**5**	**2**
BA	8	5	0	3	4	**4**	**4**
C	8	3	2	**4**	**3**	**4**	**3**
CAT	9	1	3	**4**	**5**	**4**	**5**
DD	7	3	3	6	2	**8**	**0**
DELL	10	2	1	3	2	**7**	**3**
DIS	7	4	2	**5**	**2**	**6**	**3**
EK	7	6	0	**9**	**0**	8	0
GE	9	3	1	4	5	**4**	**4**
GM	8	5	0	7	1	**8**	**1**
HD	8	3	2	**5**	**4**	**5**	**4**
HON	9	4	0	4	3	**6**	**4**
HWP	9	4	0	4	2	**8**	**2**
IBM	10	3	0	3	4	**4**	**4**
INTC	9	3	1	4	4	**4**	**3**
IP	5	4	4	**8**	**2**	**8**	**2**

Better/Worse: at least 5% higher/lower of the buy-and-hold

Experimental result (4/10)

JNJ	9	4	0	**2**	**1**	**4**	**3**
JPM	6	1	2	2	1	**3**	**0**
KO	7	4	2	**4**	**2**	4	**2**
MCD	7	3	3	**6**	**2**	6	3
MMM	8	1	4	2	3	**4**	**4**
MO	7	3	3	3	2	**6**	**2**
MRK	6	6	1	5	2	**6**	**2**
MSFT	8	3	2	4	3	**5**	**2**
NMSB	9	3	1	**12**	**0**	**12**	**0**
ORCL	8	4	1	2	4	**3**	**4**
PG	11	1	1	**3**	**1**	2	1
RYFL	4	7	0	**11**	**0**	**11**	**0**
SBC	6	5	2	7	2	**8**	**1**
SUNW	9	4	0	**7**	**4**	6	4
T	5	6	2	**5**	**5**	3	5
UTX	9	2	2	**3**	**2**	2	3
WMT	6	4	3	**5**	**2**	**6**	**3**
XOM	8	2	3	**9**	**1**	8	1
Up	282			68	83	**93**	**83**
Down		124		79	3	79	5
NC			56	29	3	30	2
Total	282	124	56	176	89	**202**	**90**

Experimental result (5/10)

RELATIVE PERFORMANCE OF ENSEMBLE APPROACHES OVER THE BUY-AND-HOLD (36 COMPANIES AND $T = 0.003$)

Symbols	market state			*Winner*		*M-Ensemble*		*A-Ensemble*		*I-Ensemble*	
	Up	*Down*	*NC*	*Better*	*Worse*	*Better*	*Worse*	*Better*	*Worse*	*Better*	*Worse*
AA	8	3	2	**7**	**3**	5	3	4	4	3	2
AXP	10	1	2	4	5	0	4	3	5	**5**	**3**
AYP	8	4	1	3	4	1	3	**3**	**3**	**2**	**2**
BA	8	5	0	5	5	2	5	4	5	**3**	**2**
C	8	3	2	4	5	2	3	**4**	**3**	2	2
CAT	9	1	3	4	6	4	5	4	6	**6**	**3**
DD	7	3	3	6	3	**6**	**1**	8	4	**7**	**2**
DELL	10	2	1	4	7	4	3	4	6	**5**	**3**
DIS	7	4	2	**6**	**3**	3	4	6	4	**5**	**2**
EK	7	6	0	7	2	6	2	7	1	**8**	**1**
GE	9	3	1	2	5	2	5	3	4	**3**	**3**
GM	8	5	0	8	3	6	2	**8**	**2**	**8**	**2**
HD	8	3	2	**2**	**4**	0	4	1	5	0	4
HON	9	4	0	**5**	**4**	4	4	**5**	**4**	2	4
HWP	9	4	0	7	3	7	2	5	3	**9**	**1**
IBM	10	3	0	3	7	3	5	4	7	**4**	**1**
INTC	9	3	1	3	6	3	6	3	7	**4**	**2**
IP	5	4	4	7	3	6	2	6	3	**7**	**0**

Experimental result (6/10)

JNJ	9	4	0	2	5	2	2	3	4	**2**	**1**
JPM	6	1	2	0	5	0	2	0	6	**2**	**2**
KO	7	4	2	4	4	2	3	**5**	**3**	3	2
MCD	7	3	3	4	5	**3**	**3**	**4**	**4**	**2**	**2**
MMM	8	1	4	2	5	1	5	2	5	**4**	**3**
MO	7	3	3	4	6	3	5	2	4	**4**	**5**
MRK	6	6	1	**7**	**2**	4	3	6	3	5	1
MSFT	8	3	2	4	4	**4**	**3**	4	4	**3**	**2**
NMSB	9	3	1	10	2	10	2	**10**	**1**	9	2
ORCL	8	4	1	4	6	5	3	3	3	**6**	**1**
PG	11	1	1	1	4	1	5	2	3	**1**	**1**
RYFL	4	7	0	**11**	**0**	**11**	**0**	**11**	**0**	**11**	**0**
SBC	6	5	2	**7**	**2**	**8**	**3**	**7**	**2**	6	2
SUNW	9	4	0	**8**	**3**	5	3	6	4	4	2
T	5	6	2	4	5	2	5	3	5	**6**	**3**
UTX	9	2	2	2	6	1	4	**2**	**3**	**2**	**3**
WMT	6	4	3	4	3	4	3	4	2	**6**	**2**
XOM	8	2	3	5	3	**7**	**2**	5	3	5	1
Up	282			57	134	52	102	62	122	**80**	**64**
Down		124		88	5	66	11	77	8	62	6
NC			56	25	9	19	6	22	5	22	4
Total	282	124	56	170	148	137	119	161	135	**164**	**74**

Experimental result (7/10)

RELATIVE PERFORMANCE OF ENSEMBLE APPROACHES OVER THE BUY-AND-HOLD (13 YEARS AND $T = 0.003$)

Year	market state			*Winner*		*M-Ensemble*		*A-Ensemble*		*I-Ensemble*	
	Up	*Down*	*NC*	*Better*	*Worse*	*Better*	*Worse*	*Better*	*Worse*	*Better*	*Worse*
1992	22	9	5	10	11	7	10	12	10	**10**	**5**
1993	21	9	6	12	12	9	11	10	7	**13**	**5**
1994	19	8	9	10	8	10	6	10	8	**9**	**3**
1995	34	0	2	5	25	4	19	6	23	**9**	**11**
1996	29	3	4	9	12	**11**	**5**	11	9	**10**	**4**
1997	32	3	1	9	15	6	11	8	14	**9**	**5**
1998	24	8	4	11	10	7	8	10	9	**10**	**6**
1999	26	6	4	16	10	14	7	15	12	**20**	**4**
2000	12	22	2	19	8	11	10	15	10	**17**	**3**
2001	12	18	5	**19**	**5**	18	5	18	5	15	4
2002	4	26	5	24	6	18	4	**22**	**3**	21	5
2003	27	5	2	11	19	12	16	12	17	**12**	**14**
2004	20	7	7	**15**	**7**	10	7	12	8	9	5
Total	282	124	56	170	148	137	119	161	135	**164**	**74**

Experimental result (8/10)

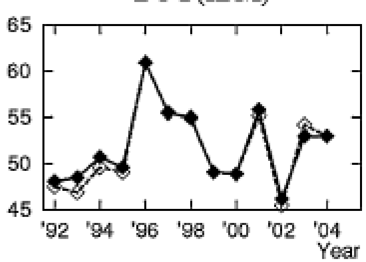
AVERAGE ACCURACY IMPROVEMENT OF GA OVER "BASE" (%)

Symbols	BUY	SELL
AA	2.96	14.92
AXP	0.99	5.50
AYP	1.30	9.01
BA	1.24	4.49
C	0.49	3.00
CAT	1.89	6.95
DD	1.43	18.27
DELL	0.60	14.67
DIS	1.04	9.26
EK	1.18	10.24
GE	0.76	17.35
GM	6.31	11.62

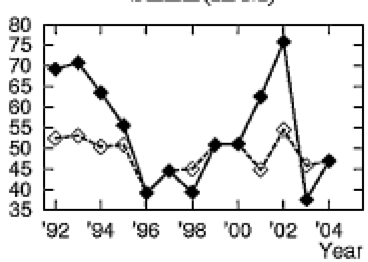
Symbols	BUY	SELL
HD	1.97	-0.02
HON	1.21	4.64
HWP	2.70	22.62
IBM	0.57	12.26
INTC	0.47	6.06
IP	1.11	7.40
JNJ	1.94	5.14
JPM	2.87	10.67
KO	1.14	8.51
MCD	1.77	5.36
MMM	2.65	8.98
MO	1.64	7.86

Symbols	BUY	SELL
MRK	1.26	13.64
MSFT	1.29	9.06
NMSB	28.82	38.42
ORCL	0.30	2.53
PG	1.83	12.37
RYFL	43.47	42.18
SBC	1.58	10.72
SUNW	0.34	8.13
T	1.30	12.57
UTX	-0.17	5.58
WMT	6.84	8.92
XOM	1.02	18.62

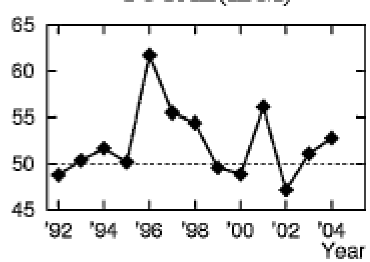
BUY(IBM)



SELL(IBM)



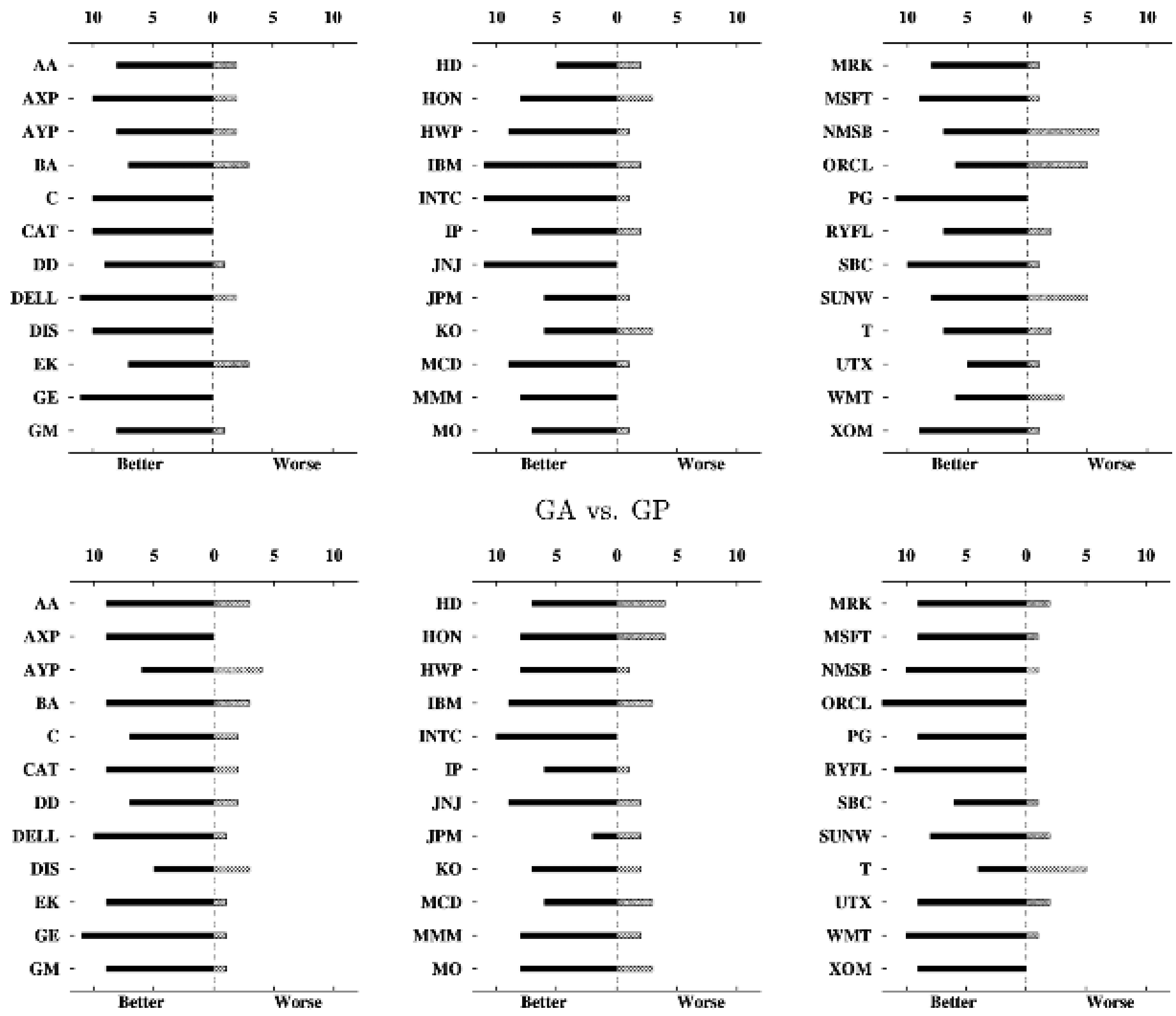
TOTAL(IBM)



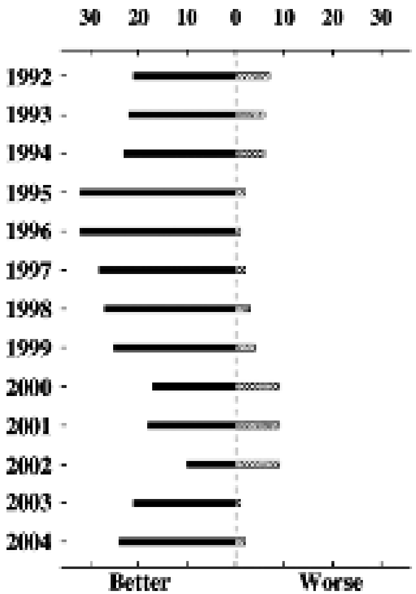
$$BASE(SIGNAL) = \frac{number\ of\ days\ of\ SINGAL}{number\ of\ days\ of\ the\ year}$$

where the ratio of the two days at least 0.1%

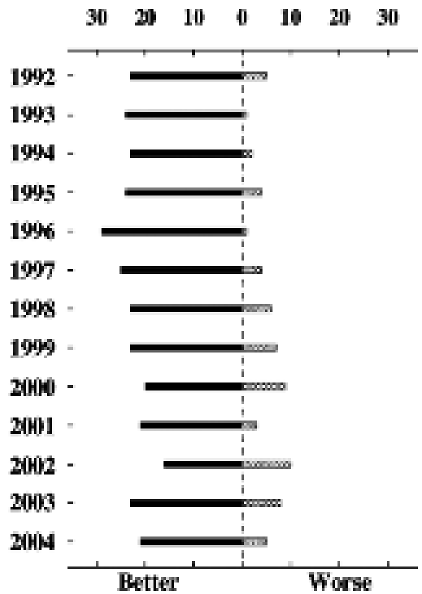
Experimental result (9/10)



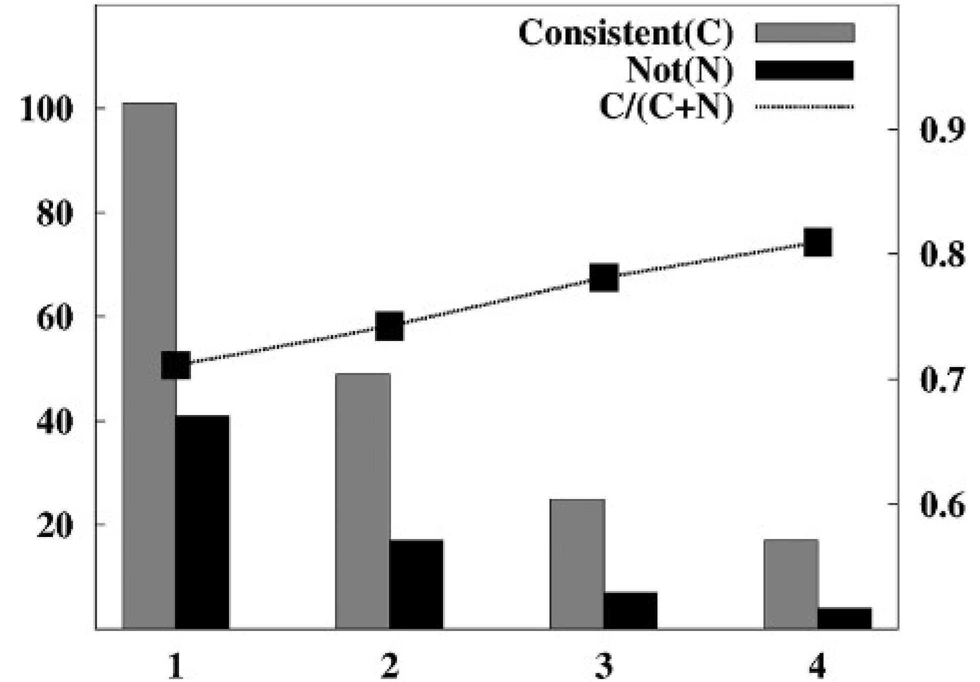
Experimental result (10/10)



GA vs. GP



GA vs. SVM



Conclusion

- Using linux cluster system with 46 CPUs using MPI, it reduced the response time about 39 times without lost performance.
- Feature work:
  - Feature selection/extraction
  - Optimize of GA